Graham Anderson

Buffalo, NY · Buffalo-area hybrid/onsite or remote
graham@grahama.co · grahama.co · linkedin.com/in/grahamanderson · github.com/grahama1970

I build agent systems that can prove what they did.

Principal AI Engineer · AI Architect · Machine Learning Engineer
Agentic AI · LLM/RAG · Knowledge Graphs · Defense & Aerospace · Founder, grahamaco

U.S. citizen. Extensive experience delivering under export-controlled (ITAR) constraints.

This is the two-page version. The longer one — full project detail and the live capability inventory — is at grahama.co/resume.

ABOUT

An unusual résumé: commercial composer for Adidas and Pepsi, Webby-recognized producer for Sony, DARPA technical lead alongside Lockheed Martin and MIT. High-end creative and hard technical work — delivered by the same person, shipped as working code, in public.

I build verifiable agentic AI systems for defense, aerospace, and compliance. On DARPA ARCOS I was Principal Data Scientist and ACERT Technical Lead on the prime-contractor team.

Self-contained by default: research, back end, front end, and the briefing. I design and build my own interfaces — React/D3 graph explorers over knowledge-graph data, agent workspaces, and grahama.co itself, all my own design and code, no agency or contractor — and I present the work myself, having briefed agentic cybersecurity research at venues across the country for DARPA ARCOS and the Air Force Research Laboratory. One person takes it from problem to running system to the room.

15+ years of hand-coding underneath it all — today I work primarily through agentic coding, driving many harnesses and models, including my own (tau). I build the tools that make agentic development trustworthy, and I use them daily. Creative systems thinking — from leading 80+ person interactive productions — is why my AI systems are architected, not stapled together.

Open to contract engagements and full-time Principal/Staff AI Engineer, AI Architect, Machine Learning Engineer, LLM Platform, and Security/Compliance AI roles.

EXPERIENCE

Founder & Principal AI Engineer / Architect | grahamaco | Buffalo–Niagara Falls Area · Remote

Feb 2025 - Present Independent AI engineering practice. Clients are primarily export-controlled (ITAR) defense/aerospace — names withheld; publicly releasable engineering is at github.com/grahama1970.

- Built tau, a receipt-gated multi-agent orchestration harness: goals compile to typed DAG contracts; every agent handoff must produce a schema-valid receipt, hashed evidence artifact, or validator result — no receipt, no action.
- Develop a heavily diverged fork of pdf_oxide (origin: yfedoseev/pdf_oxide, MIT/Apache-2.0; independent since Mar 2026): 430 commits, ~137K lines added — Rust-core changes, full Python pipeline/plugin system, PDF-cloning fixture generation, extraction calibration, and NIST document-validation tooling.
- Authored agent-skills: 340+ reusable agent capabilities (compliance evidence mapping, document AI, LLM evaluation, retrieval) and 90+ bounded worker roles, ~85% of them gated by deterministic sanity checks — public repo, private runtime.
- Built an ArangoDB agent-memory platform (private, regulated): ~219K evidence-grounded QRA records across 7K+ security-control records; hybrid BM25 + vector + multi-hop graph recall over 2.2M source chunks; ~94K Lean 4 theorem statements indexed for requirements-to-proof retrieval.
- Core contributor to pi-mono (open-source agent framework/TUI): extension system, agent runtime, and TUI harness work, Feb–Jul 2026.

Lead Research Scientist (Agentic Formal Methods) | Self-Employed / Various Clients | Buffalo, NY

Jan 2024 - Feb 2025

- Designed verifiable agentic systems: a process-driven autoformalization workflow translating engineering requirements into Lean 4 proofs for safety-critical workflows.
- Built probabilistic-deterministic self-correcting loops where LLM generation is validated by compiler/prover feedback.
- Consulted in aerospace/defense settings under export-controlled / ITAR constraints.
- Presented agentic cybersecurity research at venues across the country, including to the Air Force Research Laboratory; received an AFRL "Hacker" challenge coin. Authored the talks and decks myself.

Principal Data Scientist & ACERT Technical Lead | CS Group (DARPA ARCOS)

Sep 2020 - Dec 2023

- Joined CS Group specifically for the 4-year DARPA ARCOS program (Automated Rapid Certification of Software), on the prime-contractor team, working alongside Honeywell, Lockheed Martin, MIT, GE Research, SRI, and other program collaborators.
- Led design and delivery of ACERT (Automated Certification of Requirements Tool): architected the ArangoDB knowledge-graph schema and LLM-assisted reasoning pipeline for multi-hop compliance verification of mission-critical software against complex certification standards.
- Briefed the program and its collaborators at reviews and conferences nationally; split time roughly 50/50 between executive/technical briefings and hands-on architecture and code, authoring the decks and demos myself.

Data Scientist | grahama | NYC · Remote

Sep 2011 - Feb 2020

- Freelance data science practice across gaming, manufacturing, entertainment, education, health, and e-commerce; ITAR-rated work included.
- Clients included Toyota, Sony, Fox, Boehringer Ingelheim, UCLA Med, Dartmouth (edu-tech prediction), and Domain Industries (manufacturing intelligence).

Earlier: Interactive Executive Producer & Composer | Los Angeles, CA

2005 - 2011

- Director of Interactive Services at Dentsu America (LA division; 50% internal labor reduction, budgets $10K–$1M, teams of 3–50); Executive Producer, God of War: Ascension campaign for Sony (Webby-recognized, 80+ person productions); commercial composer for Adidas, Pepsi, and X-Games.

PUBLIC WORK (non-ITAR) — github.com/grahama1970

Client work is mostly export-controlled, so here is the public, verifiable side — including the fun stuff:

- agent-skills — my living resume: 340+ reusable agent skills, 90+ worker roles, ~85% with sanity gates. Public repo, private runtime.
- tau — receipt-gated multi-agent harness. "Agents hallucinate. Tau contains them."
- pdf_oxide — heavily diverged fork of yfedoseev's Rust PDF toolkit (430 commits, ~137K lines added: Rust-core changes, Python pipeline, PDF cloning, NIST validation).
- scillm — LLM gateway/proxy: provider routing and fallback across hosted and local models, batch pools, structured-output repair, and streaming transport for agent runtimes.
- grahama.co — this site, built and designed by me: Next.js static export, self-hosted variable type, a live d3-force capability graph, and generated surfaces that fail the build if any count drifts from the repo.
- extractor, anvil, fetcher, chatterbox voice-agent fork — supporting cast, all public.

EDUCATION

- Metis Data Science Fellowship | 2016
- Trinity University | BS, Finance, Marketing, and Economics

CORE COMPETENCIES

- Evals & Quality: LLM Evaluation, Agentic Evaluation Harnesses, Adversarial/Blind Testing, Regression Gates, Ground-Truth Fixtures, Judge Panels
- Observability & LLMOps: AI Observability, Drift Detection, Health Monitoring, Audit Trails & Receipts, Scheduled Pipelines, LLMOps
- Agentic Orchestration: Multi-Agent Systems, AI Agents, DAG Contracts, State Tracking, Tool Calling, Model Context Protocol (MCP), Bounded Agent Roles, Prompt Engineering
- LLM Platform: Large Language Models (LLM), Generative AI, LLM Gateway/Routing, Provider Fallback, Structured Outputs, Batch Inference, Guardrails
- Retrieval & Knowledge: Retrieval-Augmented Generation (RAG), GraphRAG, Knowledge Graphs, ArangoDB, Vector Databases, Hybrid BM25 + Vector Search
- Verification & Compliance: Formal Verification, Lean 4, NIST 800-53/171, MITRE ATT&CK, AI Governance
- Document AI: PDF Extraction, Layout & Table Extraction, Fixture Generation, Extraction Calibration
- Interfaces & Visualization: React, TypeScript, D3, Data-Dense & Graph UI, Design Systems, Deterministic UI Interaction Testing
- Briefing & Communication: Conference Speaking, Executive & Technical Briefings, Slide Decks & Demos, Technical Writing
- ML & Platform: Machine Learning, Model Fine-Tuning, Classifier Training, Python, Rust, Docker, Linux